

Masaki Kawamura

Student at Institute of Science Tokyo

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EDUCATION

Institute of Science Tokyo Tokyo, Japan2025 – 2027 (expected) Formerly Tokyo Intitute of Technology	Institute of Science Tokyo Tokyo, Japan2022 – 2025 Earned a Bachelor's degree one year early.	Sapporo Kita High School Hokkaido, Japan2019 – 2022
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EXPERIENCE

Research Engineer (Intern)

Turing inc. Apr 2026 – Present Tokyo, Japan
Keywords: Automonous Driving, Vision Language Action Model
Developing vision-language-action models (VLA) for autonomous driving.

Researcher

SB Intuitions Feb 2025 – Mar 2026 Tokyo, Japan
Keywords: Vision Launguage Model, Captioning
Developed dense captioning models, using reinforcement learning to improve caption quality and factual consistency.

Technical Assistant and Research Assistant

National Institute of Advanced Industrial Science and Technology (AIST) Feb 2024 – Present Tokyo, Japan
Keywords: Vision Launguage Model, CLIP
Conducting research on fine-grained CLIP models by combining object detection and caption parsing to learn detailed object features and relationships.

Backend Engineer (Intern)

Money Forward Aug 2025 (2weeks) Tokyo, Japan
Keywords: Backend, Ruby
Developed a Ruby-based backend module for an enterprise accounting system.

Research Assistant

National Institute of Infomatics (NII) Nov 2024 - Mar 2025 Tokyo, Japan
Keywords: Lauge Language Model, Synthetic Data
Fine-tuned LLMs for task and developed synthetic datasets to improve training efficiency and performance.

PROJECTS

Swallow LLM - Swallow LLM is a series of Japanese-specialized large language models developed through continual pre-training. In the development of the Swallow series, responsible for building large-scale code corpora and constructing supervised fine-tuning (SFT) datasets for reasoning tasks.

PUBLICATIONS

Masaki Kawamura, Nakamasa Inoue, Rintaro Yanagi, Hirokatsu Kataoka, Rio Yokota. PowerCLIP: Powerset Alignment for Contrastive Pre-Training. Computer Vision and Pattern Recognition (CVPR), 2026.

Kazuki Fujii, Yukito Tajima, Sakae Mizuki, **Masaki Kawamura**, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Masanari Oi, Taishi Nakamura, Takumi Okamoto, Shigeki Ishida, Kakeru Hattori, Youmi Ma, Hiroya Takamura, Rio Yokota, Jun Sakuma, Naoaki Okazaki. Rewriting Pre-Training Data Boosts LLM Performance in Math and Code. International Conference on Learning Representations (ICLR), 2026.

Taishi Nakamura, Satoki Ishikawa, **Masaki Kawamura**, Takumi Okamoto, Daisuke Nohara, Jun Suzuki, Rio Yokota. Optimal Sparsity of Mixture-of-Experts Language Models for Reasoning Tasks. International Conference on Learning Representations (ICLR) Oral, 2026.

SKILLS

Languages: *Python, Ruby, C++, JavaScript*

Frameworks & Tools: *Proficient in ML frameworks (PyTorch, NumPy)*

Research Topics: *LLM training (Continual Pretraining, SFT, GRPO, MoE), VLMs (CLIP, MLLMs), VLA*

Experienced in developing and optimizing LLMs, with a focus on Swallow models, Mixture of Experts (MoE) architectures, Test-Time Compute (TTC), and post-training techniques, such as GRPO. Skilled in both high-level VLM architectures and low-level vision-language representation learning, including CLIP. More recently, also conducting research on VLA models.